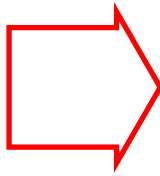


What we want:



What we have:

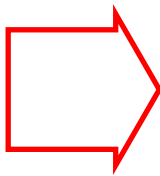
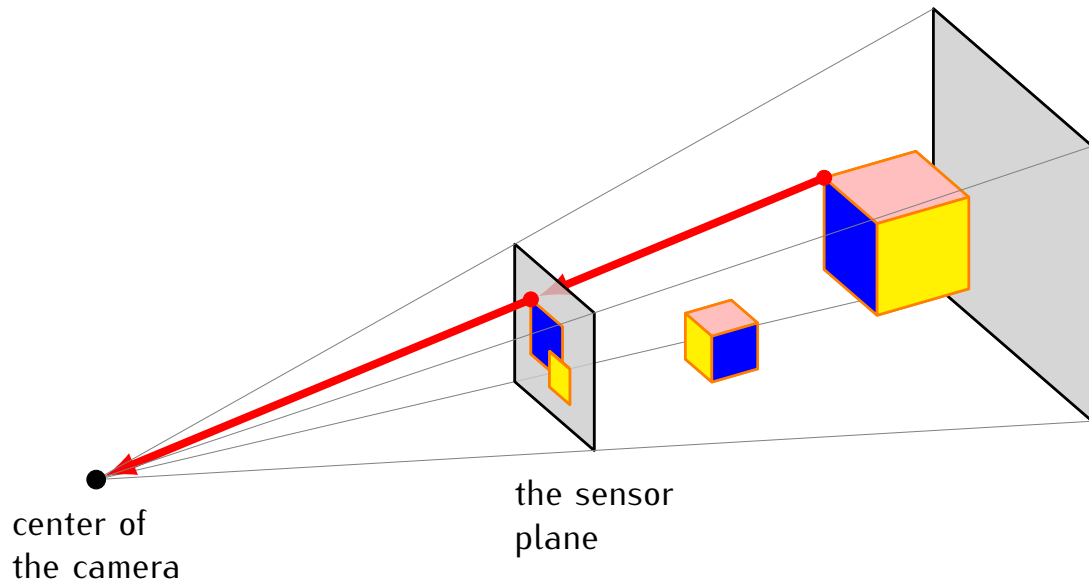
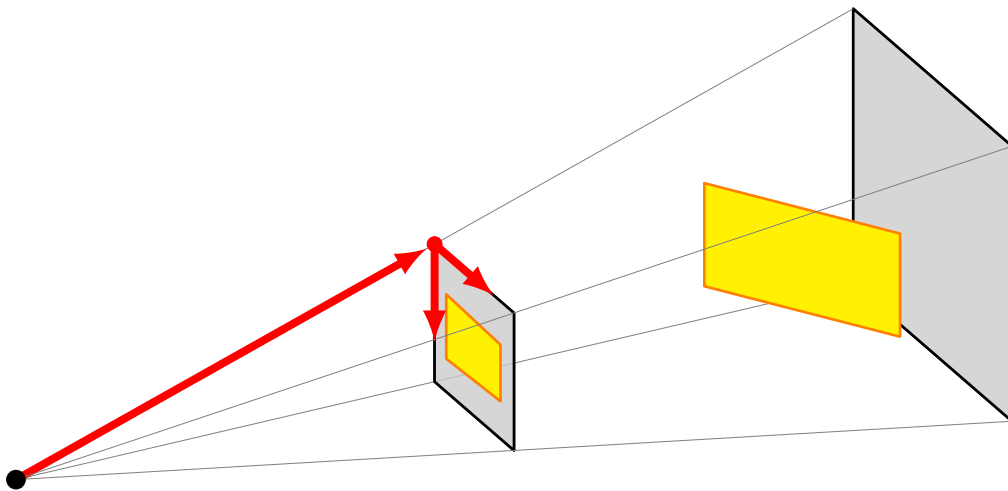
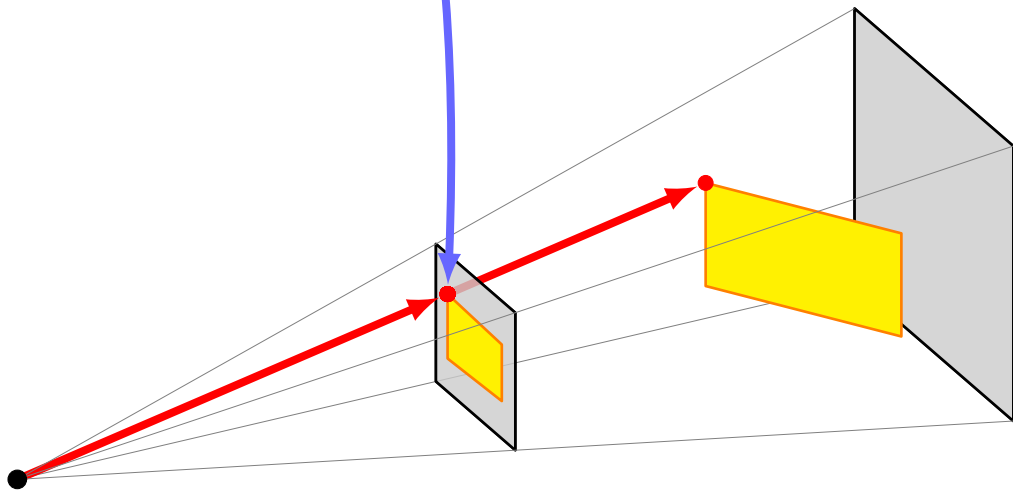


Image formation in a camera

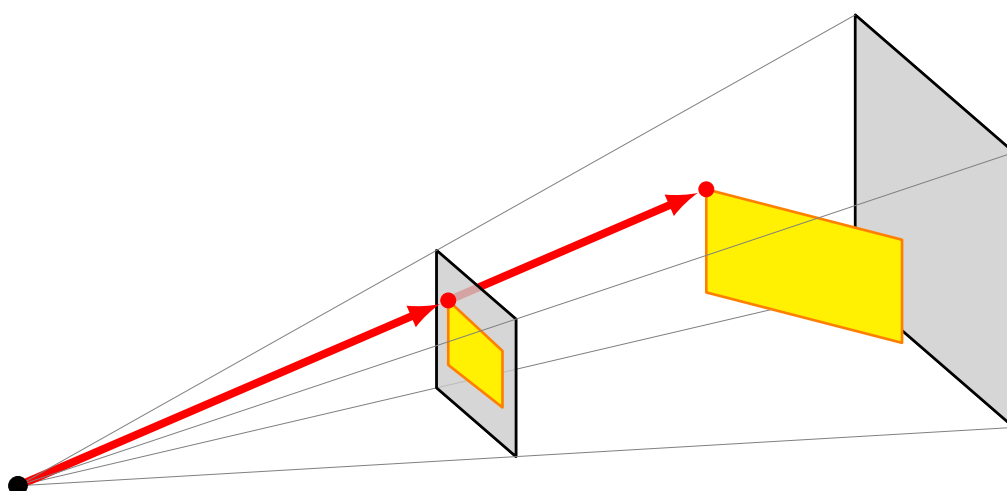
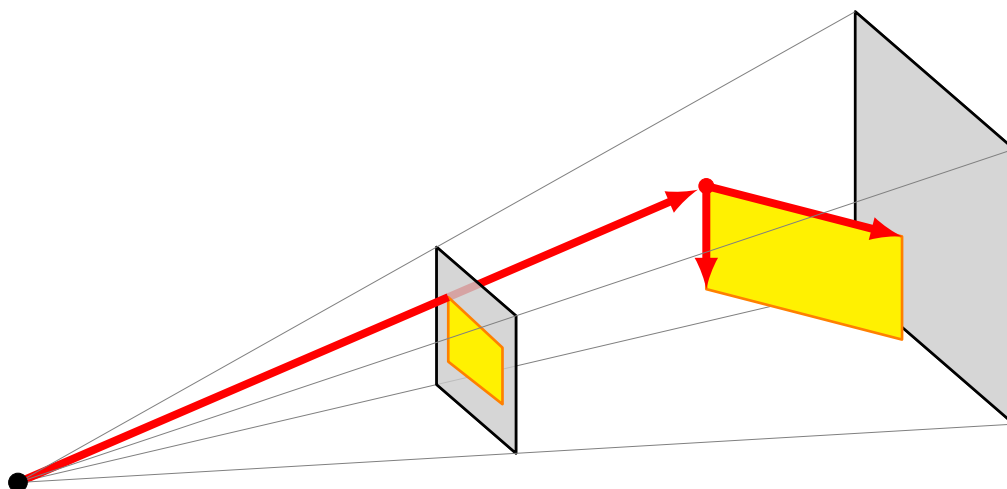


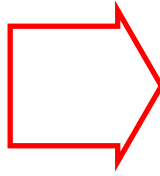
The camera coordinate system $\mathcal{C}$





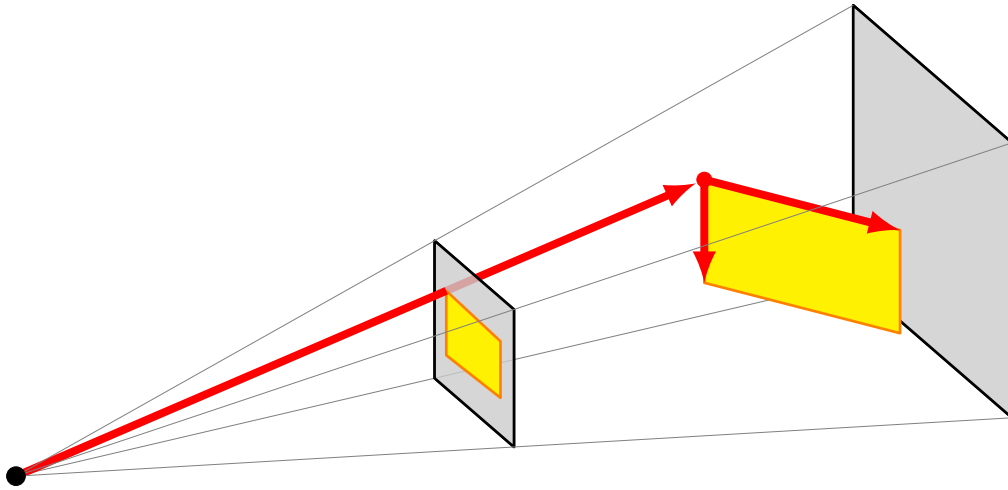
The mural coordinate system $\mathcal{M}$





From mural coordinates to camera coordinates

$$P_{c \leftarrow \mathcal{M}} = [[\mathbf{m}_1]_c \quad [\mathbf{m}_2]_c \quad [\mathbf{m}_3]_c]$$



Problem: What are the numbers a , b , c ?